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Review

Trends in marine species distribution models: a review of methodological advances and future challenges

Moritz Klaassen^{1,2,3}, Tiago A. Marques^{2,4}, Filipe Alves^{1,3} and Marc Fernandez^{1,3}

¹MARE-Marine and Environmental Sciences Centre/ARNET-Aquatic Research Network, Regional Agency for the Development of Research, Technology and Innovation (ARDITI), Funchal, Portugal

²Center for Statistics and its Applications, Department of Animal Biology, Faculty of Sciences, University of Lisbon, Lisbon, Portugal

³Faculty of Life Sciences, University of Madeira, Funchal, Portugal

⁴Centre for Research into Environmental and Ecological Modelling, University of St Andrews, St Andrews, UK

Correspondence: Moritz Klaassen (moritz.klaassen@gmail.com)

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Correlative species distribution models (SDMs) are quantitative tools in biogeography and macroecology. Building upon the ecological niche concept, they correlate environmental covariates to species presence to model habitat suitability and predict species distributions. Since their development, SDMs have undergone substantial advances in their predictive accuracy, benefiting from increased data availability, advanced machine learning algorithms, novel data integration procedures, refined model validation techniques, and incorporation of biotic predictors. Although initially applied in terrestrial systems, these models are now also widely used in the marine environment, recognized for their value in conservation planning, fisheries management, and understanding species responses to climate variability and change. Despite their increased application, SDMs face unique challenges when applied in the marine environment. These challenges include the three-dimensional complexity of marine ecosystems, the availability of environmental covariates across suitable spatial and temporal scales, the dynamic properties of these covariates, and unique dispersal patterns and mobility traits of marine species. Here, we review recent methodological advances and emerging trends in marine SDMs. We highlight three-dimensional modelling approaches that capture species distributions below the sea surface and assess the importance of temporal resolution, particularly for modelling highly mobile marine species in dynamic marine environments. Further, we discuss the expansion in the types of occurrence data being used, including fishery-dependent and fishery-independent sources, citizen science contributions, and satellite tracking data, along with the methods used to address their associated biases. We also explore and discuss novel methodologies for environmental data collection, such as remote-sensing technologies and numeric ocean models, considering the existing limitations in spatial and temporal resolution. Together, our review synthesizes methodological innovations, highlights ongoing challenges, and discusses emerging trends within the extensive literature on marine SDMs.

Keywords: correlative modelling, SDM, ENM, marine biogeography, spatial and temporal variability



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Introduction

Species distribution models (SDMs), also widely known as ecological niche models (ENMs), are numerical tools that correlate species occurrence records with environmental covariates to model the ecological niche of a species and its potential geographic distribution (Soberón 2007, Elith and Leathwick 2009, Elith and Graham 2009). Following a top-down approach, correlative SDMs operate under the assumption that the niche of a species, defined as the environmental conditions and resources required for a species to survive and reproduce, can be inferred from the species geographical presence and associated environmental conditions (but see Araújo and Guisan (2006) and Holt (2009) for detailed discussions on the niche concept in ecological models). The derived niche in the environmental space can then be projected onto geographic space, allowing for spatial and temporal inter- and extrapolation under certain assumptions (Colwell and Rangel 2009, Sillero et al. 2021, Velazco et al. 2023). Developed in the late 1970s (Zimmermann et al. 2010), these models have since evolved into indispensable tools across theoretical and applied studies in biogeography and macroecology (Guisan and Thuiller 2005, Franklin 2013, Araújo et al. 2019). They have advanced from initial environmental matching methods, such as envelope models like BIOCLIM (Busby 1991) and DOMAIN (Carpenter et al. 1993), to more sophisticated and necessarily complex approaches. These include traditional regression-based methods such as parametric models like generalized linear models (GLM; McCullagh and Nelder 1989), non-parametric models like generalized additive models (GAM; Hastie and Tibshirani 1990) and multivariate adaptive regression splines (MARS; Friedman 1991). Further, machine-learning techniques, including boosted regression trees (BRT; Friedman et al. 2000), random forest (RF; Breiman 2001), together with inhomogeneous Poisson process approaches like Maxent (Phillips et al. 2006, 2017), have become increasingly utilized in SDMs. An important addition to SDMs has been the use of model ensembles that combine predictions from multiple models, addressing the variability and uncertainty inherent in individual algorithms (Araújo and New 2007). Moreover, recent advancements in SDMs have highlighted the importance of incorporating biotic interactions, which can constitute a major driver of the ecological niche of a species, into modelling approaches (Hutchinson 1957, Soberón and Peterson 2005). This emphasis has led to the development of models that include biotic predictors in the classical SDM framework (Araújo and Luoto 2007, Fern et al. 2019) and the emerging field of joint SDMs, which model the distribution of multiple species according to environmental relationships and species associations based on their (co-)occurrences (Pollock et al. 2014, Tikhonov et al. 2017, Wilkinson et al. 2018). Further, SDMs have greatly benefited from the exponential growth in accessible high-resolution environmental datasets and species occurrence records from biodiversity repositories like the Global Biodiversity Information Facility (GBIF). The increasing amount of species occurrence data, collected not only

through structured surveys but also through citizen science (Feldman et al. 2021) and novel data collection techniques such as ancient and environmental DNA (Lagerholm et al. 2016, Riaz et al. 2020), camera traps (Gilbert et al. 2021), and acoustic monitoring (Desjonquères et al. 2022), has greatly contributed to the broader application and growing use of SDMs. One of the biggest advancements, and ongoing challenges, in SDMs is the integration of these diverse data sources into a single model, retaining the strengths of the different data sources (Miller et al. 2019, Isaac et al. 2020, Mostert and O'Hara 2023). Although these advancements in SDMs, among others, have established them as powerful tools in ecological modelling, they were primarily developed in and for terrestrial environments. As these models are now commonly applied in marine systems, it is important to explore the challenges and adaptations required for marine environments and how these have been addressed in marine SDM studies.

Challenges in the ocean

Applying SDMs to marine environments presents several challenges that stem from the inherent differences between terrestrial and marine ecosystems. Although many physical factors differ between air and seawater (Dawson and Hamner 2008, Vermeij and Grosberg 2010), influencing the physiology, behavior, and ecology of species, the primary challenge is not selecting appropriate environmental covariates, as SDMs in both terrestrial and marine contexts can be fitted with relevant covariates to model species niches. Instead, the key difficulty lies in the distinct spatial and temporal dynamics of these covariates in marine systems coupled with the high mobility of many marine species. Firstly, the ocean is inherently dynamic, with temporal variations in environmental conditions driven by various oceanographic features (Kroeker et al. 2020). Processes such as tides and currents can cause rapid, localized changes in the marine environment. In some systems, these variations are characterized by strong intra-annual fluctuations (Fig. 1a). On a larger scale, broader oceanographic phenomena like the El Niño-Southern Oscillation (ENSO) or the North Atlantic Oscillation (NAO) can induce variation of environmental conditions over longer timeframes (Fiedler 2002, Pinto and Raible 2012; Fig. 1b). This high degree of temporal variation is particularly important for SDMs, especially since many marine SDMs are applied to highly mobile and migratory species (Melo-Merino et al. 2020, Pasanisi et al. 2024). These species often traverse extensive oceanic regions such as entire ocean basins (McKeon et al. 2016). They follow not only memory (Abrahms et al. 2019a), but also changes in environmental conditions, such as prey availabilities, as they move (Corkeron and Connor 2006, Barendse et al. 2010). Secondly, the ocean's three-dimensional structure introduces complexities beyond the two-dimensional geographic spaces typically considered in terrestrial SDMs (Bentlage et al. 2013, Duffy and Chown 2017). Environmental covariates commonly used to determine species distributions, such as temperature, salinity, and oxygen, vary with depth and exhibit

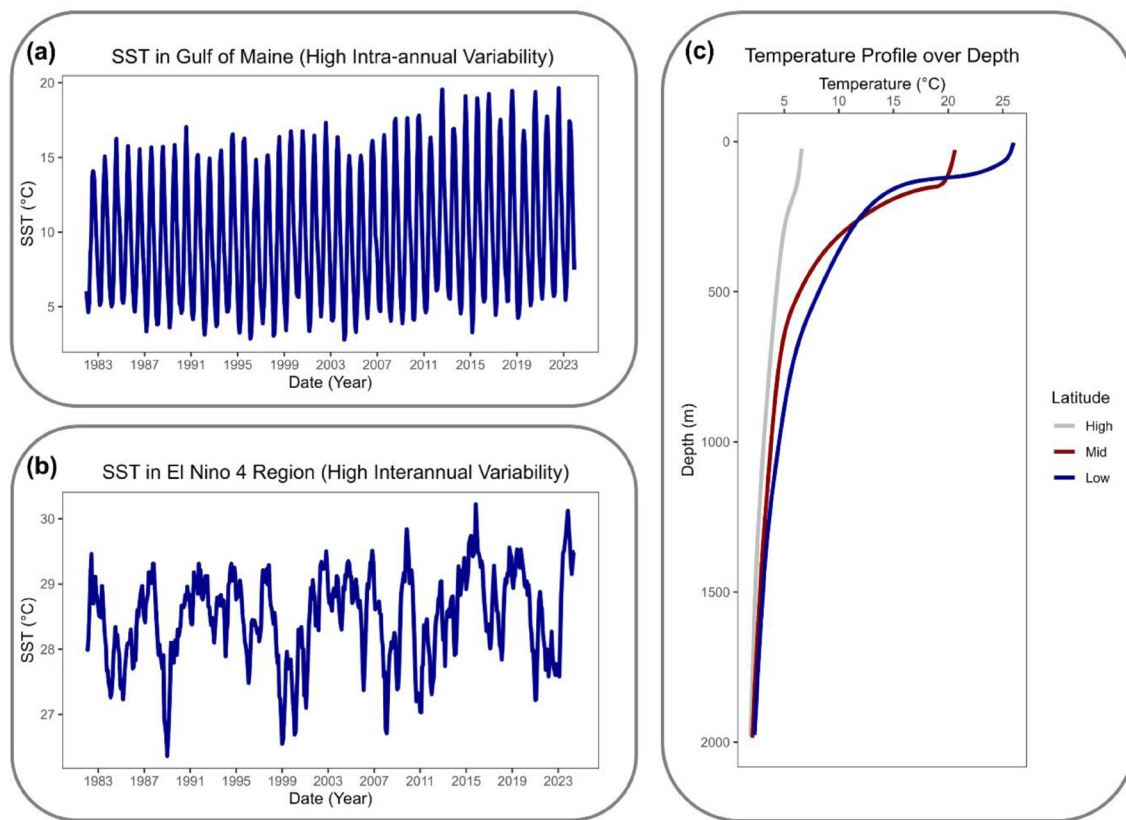


Figure 1. Spatial and temporal variability in temperature in the marine environment. (a) Temporal variability in sea surface temperature (SST) from a nearshore grid in the Gulf of Maine, displaying high intra-annual variability. (b) Temporal variability in SST in El Niño 4 region, displaying high interannual variability corresponding to El Niño and La Niña events. (c) Temperature profiles for low, mid, and high latitude regions, showing spatial variability of how temperature changes with depth. Data for this figure was sourced from NOAA OISST (Huang et al. 2020) and NOAA ARGO project (ver. 1.1.f. May 2017).

different variation patterns along latitudinal gradients. These covariates show strong variation between tropical and polar regions at the surface. However, these differences become less pronounced as depth increases, leading to more uniform environmental conditions at greater depths (Fig. 1c). In marine SDMs, the ocean's vertical dimension is important, as many species inhabit and move through varying depths and perform vertical migrations (Alves et al. 2010, Brierley 2014), encountering distinct environmental conditions such as temperature, light, oxygen, and prey availability.

Objective

Previous literature reviews of marine SDMs have given important insights into regions and taxa that have been studied, along with key ecological processes and modelling approaches that have been explored. Robinson et al. (2011) provided a foundational overview on key ecological factors such as species interactions and ontogenetic shifts and their implications for SDMs in marine environments. Robinson et al. (2017) followed with a systematic review to evaluate common practices at the time and to recommend best practices for future studies. The extensive review of Melo-Merino (2020) highlighted trends in the geographic

and taxonomic focus of marine SDMs, as well as common modelling approaches used in the field. More recently, Pasanisi et al. (2024) conducted a targeted review on cetaceans and sea turtles, providing important methodological insights for modelling these species. Therefore, with established knowledge on the regions, taxa, and algorithms used in marine SDMs, our review shifts focus to synthesizing recent advancements and addressing current challenges in the field. Specifically, this review examines three key areas of progress in marine SDMs. First, it addresses how to select the appropriate temporal resolution of environmental covariates for dynamic marine SDMs. Second, it explores methods for modelling species distributions in three dimensions. Third, it provides an overview of marine data sources, here defined as both environmental covariates (e.g. from remote sensing and ocean circulation models) and species occurrence data (e.g. from surveys and opportunistic sources). While our review does not restrict its scope to specific taxa, we highlight the importance of the temporal dimension for highly mobile species such as pelagic fish, sharks, and marine mammals, and the vertical dimension for species that inhabit distinct depth layers, such as lanternfish and jellyfish. As shown in Fig. 2, the central contribution of this review is synthesizing

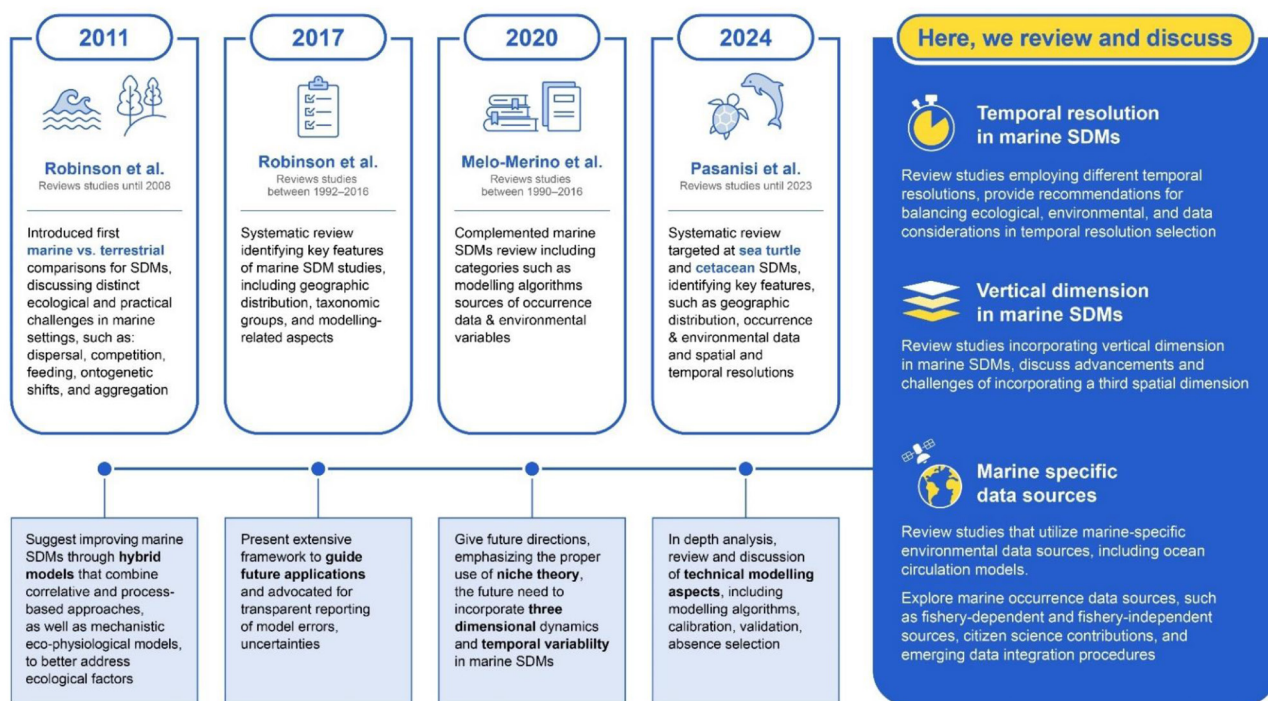


Figure 2. Overview of key literature reviews on marine SDMs. Our review connects to previous work by synthesizing recent advancements and addressing particularly three areas of progress in detail: temporal resolution, vertical dimension, and marine specific data sources.

these methodological advancements in temporal and vertical dimensions alongside emerging marine data sources, filling a critical gap by providing guidance on how careful selection of these interconnected components can improve marine SDM applications.

The temporal dimension

SDMs have traditionally been built using static covariates that lack an explicit temporal component. These static covariates include, for example, topography and substrate in terrestrial environments and bathymetry and slope in marine environments, which are constant over time (Bateman et al. 2012, Fernandez et al. 2017). On the other hand, for dynamic covariates such as temperature and precipitation, it is also common practice to use single long-term averages, typically based on a 30- or 50-year period (Milanesi et al. 2019, Stewart et al. 2021). By correlating occurrence data with these long-term averages, both the temporal aspect of the occurrence data and the inherent variability of the environmental covariates are disregarded, rendering the covariates fundamentally static as well. While this approach may be suitable to model the broad range of a species and is supported by ecological theory for stationary and sessile species with stable habitat preferences (Reside et al. 2010, Dobson et al. 2023), it might fall short, depending on the scale of the ecological question (Mannocci et al. 2017), in modelling scenarios with high environmental variability and species that are highly mobile, short-lived, or behaviorally complex (Peterson 2005, Andrew

and Fox 2020, Ingenloff and Peterson 2021). Long-term averages cannot capture environmental variability and extremes (Zimmermann et al. 2009, Perez-Navarro et al. 2022), that have been shown to largely impact ecological processes and species abundance and distributions (Bateman et al. 2012, Ghedini et al. 2015, Hill et al. 2016). By smoothing over the environmental variability within the reference period, these averages can potentially lead to biases in SDMs, or there might even be a complete temporal mismatch when the species is recorded outside the reference period of the environmental covariates (Scales et al. 2016, Abrahms et al. 2019b). Even when covariates accounting for variability, such as maximum and minimum temperatures, are incorporated into the model (Zimmermann et al. 2009), a spatiotemporal mismatch can persist between the occurrence data and environmental conditions when the species is treated as a static entity (Laube et al. 2015, Milanesi et al. 2019). This mismatch can particularly bias the model when the occurrence data precedes the reference period of covariates. To overcome these limitations, approaches to spatiotemporally match environmental data to occurrence data in SDMs have been suggested. In the literature, they are referred to as time-specific SDMs (Ingenloff and Peterson 2021), temporally explicit SDMs (Williams et al. 2017, Welch et al. 2018), and most commonly known as dynamic SDMs (Milanesi et al. 2019, El-Gabbas et al. 2021a, 2023, Dobson et al. 2023). These models temporally match the occurrence data with dynamic environmental covariates at a temporal resolution selected by the modeller (Fig. 3). By using dynamic environmental covariates, these models address the limitations of the

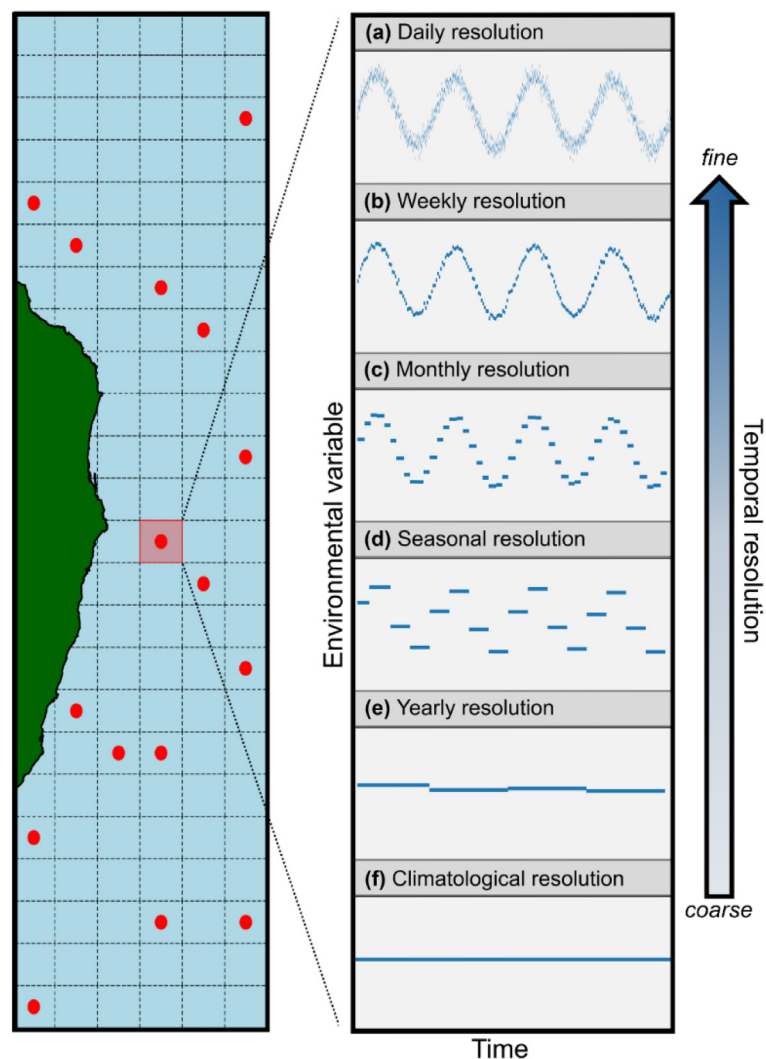


Figure 3. Example illustrating how different temporal resolutions (a–f) for the same spatial grid can smooth out environmental variability. Finer temporal resolutions preserve short-term fluctuations, whereas coarser resolutions progressively smooth out these fluctuations. Red points represent occurrence data.

equilibrium assumption in traditional SDMs, acknowledging that species distributions are not static and are responsive to changes in environmental conditions (Pearson and Dawson 2003, Milanesi et al. 2019, Foster et al. 2022). Even though major advancements have been made in developing dynamic SDMs in terrestrial environments – such as testing dynamic SDMs against static ones using virtual species (Milanesi et al. 2019), creating workflows that include temporally explicit pseudo-absence datasets (Ingenloff and Peterson 2021), and an R-package used for dynamic SDMs (Dobson et al. 2023) – their application remains relatively scarce, mainly focusing on migratory birds, short-lived insects, and species influenced by short-term environmental changes (Peterson 2005, Bateman et al. 2012, Andrew and Fox 2020, Srivastava and Carroll 2023). On the other side, as the temporal dimension becomes increasingly important for modelling highly mobile species in dynamic environments, it is unsurprising that while SDMs are applied less in marine

environments compared to terrestrial ones (Robinson et al. 2011, 2017, Melo-Merino et al. 2020), a greater proportion of marine SDMs transitioned from static to dynamic models (Andrew and Fox 2020). As shown in Fig. 3, finer temporal resolutions can better capture short-term variability, critical for highly mobile species. Further, besides the high environmental dynamism and mobility of many marine species, open ocean environments lack well-defined physical barriers for species movement and migration. Consequently, for highly mobile species that traverse extensive geographic areas, it is crucial to consider temporal resolutions in environmental covariates, which themselves can act as (dynamic) boundaries for species distributions. Dynamic marine SDMs have been applied to a range of species (Table 1), focusing on highly mobile and migratory megafauna such as cetaceans (Fernandez et al. 2018, Abrahms et al. 2019b), pelagic sharks (Hazen et al. 2018, Welch et al. 2018), sea turtles (Degenford et al. 2021), and pelagic fish such as

Table 1. Examples of studies using different temporal resolutions in dynamic marine SDMs for highly mobile species. The studies included were selected to represent the diversity of temporal resolutions, taxa and study regions in the literature. Abbreviations: BRT, boosted regression tree; GAM, generalized additive model; GAMM, generalized additive mixed model; GBM, generalized boosted model; GLM, generalized linear model; GLMM, generalized linear mixed model; MARS, multiple adaptive regression splines; PPM, Poisson point process model; RF, random forest model. Note that if authors used multiple temporal resolutions, only the finest resolution is shown in the table. Additionally, note that in most cases, the weekly resolution is provided in 8-day intervals due to the nature of satellite-derived products.

Temporal resolution	Species	Region	SDM algorithm	Reference
Daily	blue whales	California Current	BRT, GAMM, model ensemble	Abrahms et al. 2019b
	blue whales	South Pacific Ocean	BRT	Barlow and Torres 2021
	dolphins, baleen whales	California Current	BRT, GAM	Becker et al. 2020
	sea lions	California Current	GAMM	Briscoe et al. 2018
	swordfish, pelagic sharks	California Current	BRT	Brodie et al. 2018
	baleen whales	Southern Ocean	Maxent	El-Gabbas et al. 2021a
	blue whales	Weddell Sea	Maxent	El-Gabbas et al. 2023
	virtual	Azores archipelago	GBM, GLM	Fernandez et al. 2017
	swordfish, turtles, sharks, sea lions	California Current	BRT	Hazen et al. 2018
	virtual	California Current	GLMM	Scales et al. 2016
	swordfish	California Current	BRT	Scales et al. 2017
	dolphins, baleen- and toothed whales	Azores archipelago	Maxent	Fernandez et al. 2018
	blue whales	Eastern Indian Ocean	GBM	Ferreira et al. 2024
Weekly	blue whales	California Current	GAMM, BRT	Hazen et al. 2017
	baleen whales	Western Beaufort Sea	BRT, Maxent	Pendleton et al. 2020
	turtles	South-east Pacific Ocean	PPM	Degenford et al. 2021
Monthly	black marlin	South-west Pacific Ocean	Maxent	Hill et al. 2016
	dolphins, baleen whales, coastal- and pelagic sharks	South Pacific Ocean	BRT	Stephenson et al. 2023
	coastal- and pelagic sharks	Western Pacific Ocean	Maxent	Welch et al. 2018
Seasonally	baleen whales	North-East Atlantic Ocean	GAM, GBM, GLM, MARS, RF, model ensemble	Blázquez et al. 2024
	black- and blue marlin	Eastern Pacific Ocean	Maxent	Farchadi et al. 2019
	blue sharks	Western Indo-Pacific Ocean	BRT, Maxent, RF, model ensemble	Zhang et al. 2024

swordfish ([Scales et al. 2017](#), [Brodie et al. 2018](#)) and marlin ([Farchadi et al. 2019](#)). Although most studies using dynamic SDMs have focused on the Pacific Ocean, studies have also been conducted for the Arctic ([Pendleton et al. 2020](#)), Atlantic ([Fernandez et al. 2017](#)), Indian ([Ferreira et al. 2024](#)), and the Southern Ocean ([El-Gabbas et al. 2021a](#)). Different studies have utilized varying temporal resolutions in dynamic marine SDMs, ranging typically from daily to seasonal (but see exceptions where sub-daily temporal resolutions in environmental covariates have been used in marine SDMs, such as [Skov et al. 2016](#) and [Fahlbusch et al. 2024](#)). For instance, [Abrahms et al. \(2019b\)](#) modelled daily predictions of blue whales, while [Fernandez et al. \(2018\)](#) employed weekly environmental covariates for various whale and dolphin species. [Stephenson et al. \(2023\)](#) modelled the monthly probability of occurrences for dolphins, whales and sharks in the South Pacific Ocean. Additionally, [Blázquez et al. \(2024\)](#) used seasonal environmental covariates for their SDMs of baleen whales. Despite the wide use of dynamic marine SDMs, a major challenge remains the selection of an appropriate temporal resolution, which has been shown to significantly impact the model's performance. However, only a few studies have directly compared the impact of different temporal resolutions on the performance of SDMs within the same study.

[Scales et al. \(2016\)](#) found that daily or weekly resolutions of environmental data performed better than coarser resolutions such as monthly, seasonal or climatological fields in their models. [Fernandez et al. \(2017, 2018\)](#) found that weekly temporal resolutions of environmental covariates yielded better results than daily or monthly resolutions in their SDMs of virtual species and cetaceans, respectively. On the contrary, [Mannocci et al. \(2014\)](#) found that models using seasonal and climatological resolutions of environmental covariates outperformed those using weekly resolutions for cetaceans. It is important to notice that in particular marine megafauna species exhibit complex interactions with their environment on various spatiotemporal scales, ranging from daily foraging activities to extensive seasonal migrations ([Afonso et al. 2020](#)). These interactions might be most effectively detected at different temporal scales ([Fernandez et al. 2017](#)). Therefore, selecting an appropriate temporal resolution in marine SDMs is a complex task that depends on many factors.

To make an informed choice on the selection of appropriate temporal resolutions for marine SDMs, it is crucial to consider: 1) the model species behavior and ecology, 2) the degree of environmental variation, 3) the spatial scale of the study, 4) the overarching objective and application of the study, and 5) the data availability. In the following, we aim

to disentangle these factors to provide clearer guidance on selecting the most suitable temporal resolution for dynamic marine SDMs.

Ecology of the species

The choice of temporal resolution should be informed by the ecological responses of the model species to environmental variability at different temporal scales. Generally, for species whose distributions are primarily influenced by static covariates like depth, the temporal resolution of dynamic covariates may be less critical. In contrast, species that respond strongly to short-term and ephemeral environmental covariates might benefit from finer temporal resolutions. For instance, several studies have employed daily environmental covariates to model the habitat suitability of blue whales (Abrahms et al. 2019b, Barlow and Torres 2021, El-Gabbas et al. 2023). Blue whales are highly mobile animals, capable of travelling close to 100 km per day within their feeding grounds (Andrews-Goff et al. 2022, Calderan et al. 2023). They follow prey patches such as krill, which are strongly correlated with primary productivity. Primary productivity (or chlorophyll-a) are common predictors in marine SDMs, since they serve as a proxy for krill density. In that context, fine temporal resolutions of environmental covariates can serve as distal predictors of prey distributions that can largely impact the distribution of marine megafauna (Arthur et al. 2017, Barlow et al. 2020). Since the aggregation of prey patches is highly dynamic, coarse temporal resolutions of distal predictors might smooth out critical variations. On the other side, in situations of large-scale seasonal migrations, where species are expected to respond less to ephemeral features and rather to large-scale oceanographic features and derived covariates, coarse or climatological resolutions may better capture migratory patterns (Mannocci et al. 2017, Putman 2018). For example, Abrahms et al. (2019a) demonstrated that blue whale migrations in the North Pacific are influenced by long-term memory and resource tracking. They found that their movements are strongly driven by long-term climatological resource patterns, as they track the long-term average phenology of the spring phytoplankton bloom. Further, when species show a strong seasonality in their habitat utilization or a high degree of site fidelity, coarser temporal resolutions of environmental covariates might better predict their distribution (Brodie et al. 2021).

Environmental variability

The degree of environmental variability could be used to inform the selection of a suitable temporal resolution in SDMs. Generally, finer temporal resolutions may be preferred in highly dynamic systems, whereas coarser resolutions may be sufficient in more stable environments. Depending on the covariates being used, the latitudinal position of the model system might give some first insights into their variability. In systems with high temporal variations in environmental covariates, such as in temperate or polar systems, finer temporal resolutions might be preferred. For example, El-Gabbas et al. (2021a) and Pendleton et al. (2020) utilized daily and weekly resolutions of environmental covariates,

respectively, to model the distribution of baleen whales in the Southern Ocean and bowhead whales in the Arctic Ocean. In both studies, sea ice-derived covariates were important dynamic predictors. Given the high variability of sea ice (Cavalieri and Parkinson 2008), fine temporal resolutions might be preferred over coarser resolutions. In contrast, in tropical systems, where the variability in key environmental covariates like temperature and salinity is generally lower, coarser temporal resolutions may be more appropriate. Mannocci et al. (2014) found that seasonal and climatological resolutions produced better model results than weekly resolutions for cetaceans. Importantly, their study was conducted in a tropical system, where the more stable oceanographic conditions might have contributed to the effectiveness of these coarser temporal resolutions. However, while tropical systems generally exhibit lower variability in key environmental covariates, factors such as upwelling, eddies, and ocean currents can introduce significant local variability even within these more stable environments. Therefore, considering finer temporal resolutions may still be necessary when operating on a finer spatial scale. Additionally, not only the degree of environmental variability, but also the temporal scale of that variability could influence the choice of an adequate temporal resolution in SDMs. Intra-annual fluctuations driven by seasonal cycles and interannual variations linked to phenomena like the El Niño-Southern Oscillation (ENSO) can both impact species distributions, potentially requiring different temporal resolutions to be used in SDMs to accurately capture these variations. For example, Brodie et al. (2021) explored which temporal scales of environmental covariates contributed most to the predictive skill of swordfish catch and distribution. They found that the monthly climatology of environmental covariates was a good predictor in their SDMs, and that the addition of high-frequency components, such as daily variations, provided only marginal improvements in predictive accuracy. However, they found that adding a low-frequency component that captured interannual variations, improved the predictive skill in their models. Similarly, González García et al. (2018) demonstrated the importance of using different spatiotemporal resolutions to account for different oceanographic processes, particularly in highly dynamic marine environments. Their multi-scale analysis of blue whale habitat preferences surrounding the Azores highlights how both long-term events, like seasonal blooms, detected by coarser temporal resolutions, and short-term features, like eddies and fronts, detected by finer temporal resolutions, can both influence species distributions.

Spatial scale

Given the connection between the spatial scale, and temporal variability of environmental covariates, care should be taken in selecting appropriate temporal resolutions depending on the spatial scale of the study (Mannocci et al. 2017, Brodie et al. 2021). Generally, finer spatial scales are associated with higher temporal variability (Fernandez et al. 2021). At these fine scales, species are often influenced by highly dynamic environmental features derived from small-scale

ocean currents, tides, eddies, or prey aggregations, which can change quickly (Johnston et al. 2005, Fahlbusch et al. 2022). Consequently, on fine spatial scales, fine temporal resolutions of environmental covariates, such as daily or weekly fields, may best describe these rapidly changing environmental conditions that drive fine-scale species distributions (Fernandez et al. 2017). In contrast, environmental variability tends to be more stable at larger spatial scales, such as across entire ocean basins or large migratory corridors. At these scales, species distributions may be driven by rather large-scale oceanographic phenomena, which may not fluctuate significantly over short time periods (Farchadi et al. 2019). As a result, coarser temporal resolutions, such as monthly or even seasonal averages, might be more appropriate. Further, this approach can smooth out short-term variability, which may be less relevant to species distributions at coarse spatial scales. For instance, Becker et al. (2010) demonstrated that the coarsest spatial sea surface temperature (SST) resolutions had the highest predictive performance for most of the cetacean species in their models. This improved performance was attributed to several factors, including the reduction of data gaps at coarser resolutions and smoothing out fine-scale variability that might introduce noise into models.

Objective

The application and general objective of the study should also be taken into consideration to make an informed choice of adequate temporal resolutions in SDMs. For example, when the SDM aims to understand the spatial distribution and habitat suitability of a species without considering the temporal aspects of that distribution, coarse temporal resolutions or even static approaches using climatological averages of environmental covariates may be sufficient. However, when the objective of the SDM is related to the distribution of the species across both space and time, dynamic SDMs are naturally more suitable. For instance, Hill et al. (2016) used SDMs to track distributional shifts of a pelagic predator in the South-west Pacific Ocean over time, calculating a poleward shift by employing a monthly temporal resolution of covariates across a 16-year period. Similarly, Farchadi et al. (2019) examined how the spatial distribution of suitable habitats for their model species varied seasonally and in response to ENSO events using seasonal fields of their environmental covariates. In these examples, coarse temporal resolutions of months and seasons, respectively, were effective to understand broad-scale distributional shifts of suitable habitats. However, in conservation and management applications, particularly within dynamic ocean management, employing finer temporal resolutions in marine SDMs may be more suitable. In these approaches, where management strategies change dynamically in space and time responding to changes in species distribution (Maxwell et al. 2015), SDMs with fine temporal resolutions can be used for a myriad of objectives, like estimating ship strike risks (Hazen et al. 2017), reducing bycatch (Hazen et al. 2018), estimating effectiveness of marine protected areas (Welch et al. 2018), and fisheries sustainability (Scales et al. 2017) in near real-time.

Data availability

Building SDMs at fine temporal resolutions requires that both species occurrence data and environmental covariates are available at these resolutions. However, in practice, many oceanographic covariates are often unavailable at fine temporal resolutions in many ocean regions, and many datasets are restricted to surface covariates only (Fernandez et al. 2017, El-Gabbas et al. 2021b). Additionally, it is common to find discrepancies where only a limited number of covariates are available at high temporal resolutions, while others are accessible only at coarser scales. This inconsistency can introduce challenges, as modelling across multiple temporal resolutions may complicate the predictions of the model. When only a few covariates are available at the desired temporal resolutions, omitting key covariates that are unavailable at these resolutions could result in less accurate models, as these covariates may be crucial in defining species niches. Further, there may be a trade-off between achieving finer temporal resolutions and the quality of environmental covariate data, which can be compromised by data gaps due to cloud cover in many remote sensing products (Viña et al. 2008). This issue has been extensively discussed in the marine literature, with early considerations of the limitations posed by cloud cover provided by Smith et al. (1986), followed by discussions from Redfern et al. (2006) and Becker et al. (2010). The key challenge is choosing between fine temporal resolutions, such as daily satellite images, which provide detailed dynamic features but are often interrupted by data gaps due to cloud cover, or coarser resolutions, which fill in these gaps but may smooth over dynamic details. While the issue of cloud cover in remote sensing products is commonly known today for researchers in the field, Scales et al. (2016) showed through data simulation that their model performance declined substantially when cloud cover caused more than 70% of data loss in environmental covariate data. Therefore, in regions with overall higher cloud cover, careful consideration of data loss in high-resolution environmental covariates should be given when using remote sensing products.

Balancing the selection of temporal resolutions

Determining the appropriate temporal resolution for dynamic marine SDMs is a complex task that should not be made in isolation but instead consider a combination of the ecology of the species, the environmental dynamism, the spatial scale of the study, the availability and quality of environmental data, and the specific objectives of the research. In cases where environmental dynamics suggest fine temporal scales, a coarser resolution may be adequate if the species is relatively stationary, particularly in regions with frequent cloud cover where the focus is to model the broad habitat suitability of a species. Conversely, in more stable environments where coarser scales might be sufficient, finer temporal resolutions may be necessary for highly mobile species, especially when cloud-free datasets or other modelled products are accessible, and the study aim is to support dynamic ocean management. Lastly, while marine SDMs have been applied at very fine spatiotemporal scales, it is important to

recognize that as we move towards these fine resolutions, we may transition from the realm of SDMs into the domain of animal movement analyses, where different and potentially more appropriate statistical approaches exist, such as resource and (integrated) step selection analysis (Avgar et al. 2016, Signer et al. 2017, Fieberg et al. 2021), continuous time movement modelling (Calabrese et al. 2016), and state-space models (Patterson et al. 2008).

Modelling in three dimensions

The vertical dimension of the ocean introduces a variety of habitats and ecological zones characterized by distinct physical and biological properties (Gregory 2000, Sammartino et al. 2018). As depth increases, strong gradients, including thermoclines (temperature), haloclines (salinity), and oxyclines (oxygen), create distinct layers with varying environmental conditions, shaping microclimates within the vertical water column (Robinson et al. 2011). These abiotic changes along the vertical gradient are reflected in the evolutionary adaptations and ecological preferences of different pelagic species (Gaither et al. 2016, Baalsrud et al. 2017) and influence their vertical distribution (Sutton 2013). Further, the vertical distribution of marine species is not static; many marine species, including pelagic zooplankton, fish, and squid, perform daily vertical migrations, moving between different depth zones in response to environmental cues such as light and food availability (Brierley 2014, Bandara et al. 2021). Also, marine megafauna species such as cetaceans, sea turtles, and large fish dive to various depths in response to prey availability, predator avoidance and physiological maintenance. Together, the vertical space use plays a crucial role in ecosystem dynamics, redistributing biomass and energy throughout the water column and influencing predator-prey interactions across depth gradients (Alves et al. 2010, Kelly et al. 2019, Pinti and Visser 2019). Many ecological models increasingly acknowledge the importance of integrating the vertical habitat structure of marine environments. For instance, advances in animal-borne telemetry tags have enabled several studies to incorporate vertical space use in animal movement models. These models explore the connections between vertical space use and various factors, including behavioural states (Bestley et al. 2015, Griffiths et al. 2018), environmental covariates (Barbour et al. 2023), and prey densities (D'Antonio et al. 2024). Moreover, the vertical dimension of the marine environment has been utilized in spatial models of abundance and density (Manderson et al. 2011, Astarloa et al. 2021), habitat preference (Photopoulou et al. 2013) and is becoming an important consideration for marine conservation (Venegas-Li et al. 2017, Doxa et al. 2022, 2025).

Since early discussions by Dambach and Rödder (2011), there has emerged a consensus on the importance of integrating the vertical dimension into marine SDMs, which has been recognized as one of the major challenges in the field (Melo-Merino et al. 2020). In terrestrial SDMs, the environmental data inherently includes elevation and therefore,

already operates on a vertical spatial scale of the occurrence data (with exceptions for avian habitats, see Koma et al. (2022) for fine-scale inclusion of vertical habitat structure in SDMs). On the contrary, marine environmental data does not inherently match the species occurrence data at depth. This disparity has led to the development of diverse methods to conceptually and practically integrate the vertical dimension into marine SDMs. When depth information of the occurrence data is absent, the SDM is essentially 2D, which may be appropriate for benthic species with a narrow depth distribution (Duffy and Chown 2017, Simões et al. 2021). Generally, 2D SDMs can include sea-surface covariates only; however, it is becoming more common to couple these with subsurface covariates. These subsurface covariates can be categorized into static bathymetry-derived covariates such as seabed rugosity, slope and ocean depth, and dynamic simplified at-depth covariates such as the temperature at the bottom of the water column or at specific depths (Duffy and Chown 2017, Chen et al. 2021, Tang et al. 2024). Even though depth has been shown to be an important covariate in many marine SDMs (Stirling et al. 2016, Hazen et al. 2018, Burgos et al. 2020, Guillaumot et al. 2021), it might act as a confounder and cause collinearity issues when functioning as a distal predictor, influencing predictions indirectly through other covariates (Elith and Leathwick 2009). On the other side, in cases such as for deep-diving cetaceans, depth can act as a geographic limiting factor (Silva et al. 2014, Tobeña et al. 2016) and directly influence their distribution, thereby functioning as a proximal predictor (Fernandez et al. 2017, 2018). Further, oceanographic subsurface covariates have been evaluated in marine SDMs. For instance, Brodie et al. (2018) showed that isothermal layer depth and bulk buoyancy frequency improved the explanatory power and predictive performance of most of their SDMs of pelagic predators. Additionally, covariates such as upper ocean heat content and eddy kinetic energy (Abrahms et al. 2019b, Scales et al. 2023) and mixed layer depth (Erauskin-Extramiana et al. 2019) have been utilized in marine SDMs. Tang et al. (2024) tested the impact of depth of covariates such as temperature, salinity, oxygen levels, and primary production on the performance of SDMs for pelagic fish. They found that the most accurate model fit was achieved when the depths of these covariates were consistent with the actual depths at which the model species are typically found. Despite the incorporation of subsurface covariates in marine SDMs, the models remain fundamentally 2D when depth is not assigned to the species occurrence data. When the occurrence data includes depth information and is assigned to the nearest of multiple (discrete) vertical depth layers as chosen by the modeller, the model becomes quasi-three-dimensional. This method, conceptualized as 2.5D SDMs by Duffy and Chown (2017), involves fitting the SDM for the same species at different depths (Fig. 4). These models have been applied to various pelagic species across geographic regions. Studies implementing 2.5D SDMs have found promising results for species with a known affinity for deeper habitats (Freer et al. 2020). Additionally, 2.5D SDMs enabled

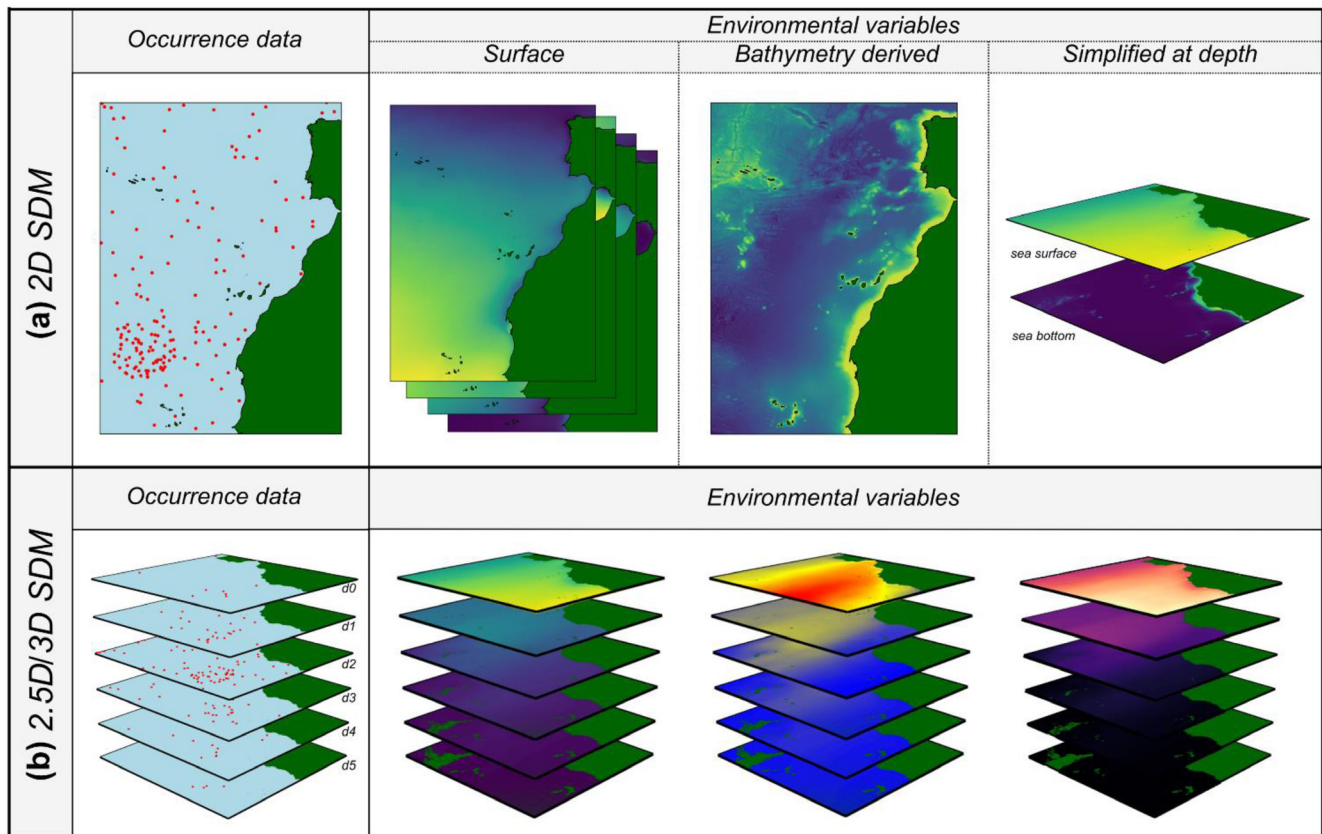


Figure 4. Examples of spatial matching of occurrence and environmental data. (a) 2D SDM, where the occurrence data is matched with surface-, bathymetry derived- or simplified at depth covariates. (b) 2.5/3D SDM, where the depth of the occurrence data (in this example denoted by d0-d5) is explicitly matched to the depth of the environmental covariates. Data for this figure was sourced from Bio-Oracle (Assis et al. 2018) and World Ocean Atlas (Garcia et al. 2019).

the exploration of diversity across depths (Pantiukhin et al. 2023), macroecological patterns such as the isothermal distribution of marine fish (Valle et al. 2024) and variations of occurrence across different depths over time (Andrews et al. 2020). Another method of incorporating the vertical dimension into marine SDMs was introduced by Bentlage et al. (2013). Their approach transforms environmental covariates and species occurrence data across various depth layers into a two-dimensional space for modelling, followed by a back-transformation to interpret the models in three-dimensional contexts. This method has been effectively utilized in marine SDMs, such as by Verhaegen et al. (2023), who showed that the distribution of different morphotypes and genetic clades of midwater jellyfish was mainly explained by depth. Further advancements have introduced explicit 3D SDMs. These models match the three-dimensional environmental data to the specific depths recorded for each species occurrence and, as such, treat depth continuously rather than as discrete layers. The recently developed R-package 'voluModel' (Owens and Rahbek 2023, www.r-project.org) facilitates the workflow of such 3D SDMs by extracting environmental data for 3D occurrences. By providing tools for down-sampling occurrence data in 3D, interpolation of environmental data, and assessing extrapolation risk in 3D (Elith et al. 2010),

'voluModel' enhances the accessibility and reproducibility of marine SDMs in 3D. Despite progress in 3D approaches to marine SDMs and the evident necessity to incorporate covariates reflecting the actual depth of marine species, especially for pelagic species that inhabit depths below the sea surface, some concerns about 3D SDMs persist. One of these concerns is the availability of occurrence data since not all entries in common databases, such as the Global Biodiversity Information Facility (GBIF) and Ocean Biogeographic Information System (OBIS), include depth. Moreover, even when they include depth information to the species occurrence data, it is often unclear whether the data represents the maximum depth reached by the net or the actual depth at which the species was caught (Freer et al. 2020). This raises critical considerations of the trade-off between data quantity and quality, leading some studies to opt for 2D models instead of 3D models to address these concerns (Woods et al. 2023). Further, environmental covariates at specific depths, as provided by sources like Marine Copernicus and the World Ocean Atlas (WOA), are often interpolated, which raises the possibility of spatial and temporal inaccuracies, which could influence the predictive accuracy of the models that use them. Among the studies identified in this review (Table 2), those that integrate the vertical dimension into their SDMs (2.5D

Table 2. Examples of studies using different approaches to integrate the vertical dimension in marine SDMs. The studies included were selected to represent the diversity of taxa, study regions, and methods for integrating the vertical dimension into marine SDMs. Abbreviations: AOU, apparent oxygen utilization; BBV, bulk buoyancy frequency; BRT, boosted regression tree; Chl, chlorophyll a concentration; CTA, classification tree analysis; DO, dissolved oxygen; D20, depth of the 20°C isotherm; EKE300, eddy kinetic energy in upper 300m; FDA, flexible discriminant analysis; GAM, generalized additive model; GARP, genetic algorithm for rule-set prediction; GBM, generalized boosted model; GLM, generalized linear model; HC300, heat content in upper 300m; ILD, isothermal layer depth; MARS, multiple adaptive regression splines; NOO, niche of occurrence; NPP, net primary production; NPPEN, non-parametric probabilistic ecological niche model; RF, random forest model; SBT, sea bottom temperature; SC-GAM, shape-constrained generalized additive model; U100, meridional velocity at 100m depth; V100, zonal velocity at 100m depth. Note that only subsurface covariates are listed, even though studies might have coupled them with surface covariates.

Depth integration	Species	Region	SDM algorithm	Subsurface covariates	Covariates source	Reference
2D SDMs with subsurface covariates	swordfish, pelagic sharks	California Current	BRT	BBV, ILD	Regional Ocean Modeling System (ROMS)	Brodie et al. 2018
	billfish, tuna	South-west Pacific Ocean	BRT	D20, EKE300, HC300, MLD, Temp500, U100, V100	CSIRO Climate Analysis and Forecast Ensemble system (CAFE60)	Scales et al. 2023
	cephalopods	European Seas	ANN, FDA, GAM, GBM, GLM, , MARS, NPPEN, RF, model ensemble	SBT, SBT range, SBT variance	CORA: Coriolis Ocean database for ReAnalysis	Schickele et al. 2021
	swordfish	West Indian Ocean	Maxent, RF	At maximum depth of 500m in 100m intervals: DO, NPP, salinity, temperature	Copernicus Marine Service	Tang et al. 2024
2.5D/3D SDMs	capelin	North-west Atlantic Ocean	Maxent	At 47 depth layers (~ 0.5–~1045 m): Chl, oxygen, salinity, temperature	GLORYS V4.1 BIOMER V3.2	Andrews et al. 2020
	deep-sea jellyfish	Global open ocean	Maxent, GARP	At 24 depth layers (0–5750 m): DO, nitrate, percent oxygen saturation, percent oxygen utilization, phosphate, salinity, silicate, temperature	World Ocean Atlas 2005 (WOA05)	Bentlage et al. 2013
	pipefish, lanternfish, eel, deep-sea fish	North-east Atlantic	ANN, BIOCLIM, BRT, CTA, FDA, GAM, GLM, Mars, Maxent, RF, model ensemble	At 96 depth layers (0–4900 m): DO, temperature, salinity	World Ocean Atlas 2013 (WOA13)	Duffy and Chown 2017
	lanternfish	Eastern coast of Australia	GAM, Maxent	At 3 depth layers (0, 150, 500 m): DO, nitrate, phosphate, temperature, salinity	CSIRO Atlas of Regional Seas (CARS) 2009	Flynn and Marshall 2013
	lanternfish	Southern Ocean	Maxent	At 7 depth layers (0–1000 m): DO, temperature, salinity	World Ocean Atlas 2013 (WOA13)	Freer et al. 2020
	luminous hake	Caribbean Sea, Gulf of Mexico	GLM	5–800 m: AOU, temperature	World Ocean Atlas 2018 (WOA18)	Owens and Rahbek 2023
	commercial fish species	Atlantic Ocean	SC-GAM	At 47 depth layers (0–1000 m): DO, nitrate, NPP, temperature, salinity	World Ocean Atlas 2018 (WOA18) Copernicus Marine Service	Valle et al. 2024
	midwater jellyfish	Global	Maxent, NOO	At 64 depth layers (50–2700 m): DO, salinity, temperature	World Ocean Atlas 2018 (WOA18)	Verhaegen et al. 2023

or 3D), specifically model pelagic midwater and deep-sea species. This prompts the question of whether and how the vertical dimension in SDMs can accommodate species that exhibit significant vertical migrations and whose positions in the water column are highly time dependent.. Generally, selecting between 2D, 2.5D, and 3D SDMs should be guided by the vertical resolution of both the occurrence and environmental data, as well as the known depth-related behaviors of the species. A 2D SDM may suffice for benthic species or when depth data are entirely missing. If the species exhibits depth-specific behavior and occurrence data includes coarse depth classes (e.g. net tow depths), a 2.5D SDM – where the model is fitted across discrete depth layers – offers a practical compromise. However, when high-quality depth-resolved occurrence data are available and can be matched to high-quality environmental layers at depth, fully 3D SDMs become appropriate.

Marine data sources

Environmental covariates

Early studies modelling marine species distributions primarily relied on environmental data collected in situ at the time of animal observation (Hedley et al. 1999, Forney 2000, Hooker et al. 2001). This would prevent models from being used at spatial and temporal scales beyond those where the data was collected, hindering the applicability and transferability of models. However, advances in satellite remote sensing at the time have demonstrated that key surface covariates, such as SST and chlorophyll-a, can be effectively captured through satellite sensors (Casey and Cornillon 1999, Gregg and Casey 2004). The ability of satellite sensors to provide synoptic coverage of broad ocean areas on a frequent and real-time basis has since established them as a major source of environmental covariates used in ecological modelling (see He et al. (2015) for a comprehensive review of remote sensing in SDMs).

Today, environmental covariates for marine SDMs are most commonly sourced from global datasets that contain both in situ collected data and data from high-resolution satellite sensors. In situ data is now collected through a wide range of methods, including sonar-based surveys, buoys, ships, and Argo floats, providing greater spatial and temporal coverage beyond the location and time of occurrence data collection. Satellite data is typically used to obtain broad-scale covariates such as SST, chlorophyll-a, and sea ice extent, while in situ data provides covariates at depth and covariates that are challenging or not possible to measure via satellite remote sensing such as pH levels, nutrient concentrations, and dissolved oxygen. Some of the most widely used portals for datasets include Bio-Oracle (www.bio-oracle.org), Copernicus Marine Environment Monitoring Service (www.copernicus.eu/en/copernicus-services/marine), MARSPEC (www.marspec.org), the Ocean Color Web (<https://ocean-color.gsfc.nasa.gov>), and the World Ocean Atlas ([www.ncei.](http://www.ncei.noaa.gov/products/world-ocean-atlas)

[noaa.gov/products/world-ocean-atlas](http://www.ncei.noaa.gov/products/world-ocean-atlas)). Frequently updated, these databases provide a wide range of covariates that can be used in marine SDMs. In recent years, marine SDMs have increasingly incorporated environmental data from configurations of numerical models, particularly from physical ocean circulation models such as the hybrid coordinate ocean model (HYCOM; Becker et al. 2022), nucleus for european modeling of the ocean (NEMO; Fernandez et al. 2021), or the regional ocean modeling system (ROMS; Karp et al. 2022). These mechanistic models are sources for essential environmental covariates, such as SST and salinity, and covariates that reflect dynamic ocean processes such as eddy kinetic energy and current velocities, which have been previously difficult to obtain. These sources are especially advantageous for fine-scale SDM applications, as they provide environmental covariates on high spatial and temporal resolutions (Fernandez et al. 2021), which are particularly important for dynamic SDMs. For example, Becker et al. (2020) utilized daily environmental data derived from ROMS to build their SDMs of dolphins and baleen whales in the California Current. In particular dynamic SDMs benefit from the continuous environmental data provided by ocean circulation models since they are unaffected by cloud cover, which often causes data gaps in traditional satellite remote sensing products (Scales et al. 2017). Additionally, these sources provide information on the vertical ocean dimension, capturing subsurface features such as mixed layer depth and water column stability, that have been shown to improve the predictive performance in marine SDMs (Brodie et al. 2018). While only a few studies have directly compared the performance of SDMs based on environmental data from ocean circulation models to those using traditional in situ and satellite data, the existing studies have shown promising results. For example, Becker et al. (2016) demonstrated that their models based on ROMS data could reliably predict cetacean abundance in the California Current, achieving comparable accuracy to models built with in situ data. Similarly, Scales et al. (2016) found that models utilizing ROMS data performed comparably to those based on satellite remote sensing and recommend using ROMS-derived environmental data, particularly for SDMs on fine temporal resolutions for dynamic management applications. Together, with fine and consistent spatiotemporal resolutions among environmental covariates, not being affected by cloud cover, and providing detailed information on the vertical ocean dimension, ocean circulation models present a promising source of environmental data for improving marine SDM performance. Nonetheless, care is required to further understand when and where the data from such mechanistic models might be relevant – which might be model dependent – and at what spatiotemporal scales it can be reliably used. Besides physical ocean models, also numerical biogeochemical and ecosystem models have been used as sources of predictor covariates for marine SDMs. For instance, Pérez-Jorge et al. (2020) applied an ecosystem and population dynamics model to incorporate prey-related covariates into SDMs for baleen whales in the mid-North Atlantic Ocean.

Occurrence data

Fishery-dependent and fishery-independent data

Although often accessed through regional or global databases, the original sources of marine species occurrence data can be categorized into fishery-dependent or fishery-independent sources (Pennino et al. 2016). Generally, the occurrence data collection in marine environments poses unique challenges compared to terrestrial settings due to the difficulty of accessing many oceanic regions, the high costs associated with dedicated surveys, and the limited applicability of traditional citizen science initiatives (Alves et al. 2023). Therefore, occurrence data for many marine species and regions used in SDMs often relies on fishery-dependent data sources (Karp et al. 2022, Rodrigues et al. 2022). Fishery-dependent data are inherently tied to fishing operations and include catch data on the target species, which is often detailed in logbooks, observed catch landings or documented by on-board fishery observers. Moreover, depending on the legislation, fishers are often required to report bycatch data, which refers to the incidental capture of non-target species (Basran and Sigurðsson 2021). Even though concerns regarding under-reporting exist, the bycatch data is also an important component of fishery-dependent data used in marine SDMs. For example, Lezama-Ochoa et al. (2020) used bycatch data from the tropical tuna purse-seine fishery to model the distribution of devil rays in the eastern tropical Atlantic Ocean. While fishery-dependent data are abundant, cost-effective, and often provide broad temporal coverage, through non-random and preferential sampling, they are spatially biased to areas where fishing occurs (Ducharme-Barth et al. 2022). Besides the presence-only nature of this data in most instances (Pan et al. 2023), this spatial bias is the major drawback of fishery-dependent data posing a risk of overfitting to environmental covariates found in the fishing grounds (i.e. niche truncation) (Araújo and Guisan 2006, Chevalier et al. 2022). In contrast, fishery-independent data sources, gathered through scientifically designed surveys offer a systematic approach, collecting both presence and absence data of the species through a standardized sampling design (but see Fernandez et al. (2022) for discussion on absences in SDMs of highly mobile marine species). Although more expensive, logistically challenging and offering a shorter temporal coverage, fishery-independent data provide a more spatially unbiased representation of species occurrences in marine environments. While models based on fishery-dependent data are effective at identifying temporal relationships, and fishery-independent data provides more precise spatial predictions (Pennino et al. 2016), integrating these data types to combine the strengths of each within a single model is currently a major focus of ongoing research (Isaac et al. 2020). For example, Alglave et al. (2022) developed a spatial hierarchical model that integrates both scientific survey data and commercial catch data, specifically addressing the challenge of preferential sampling in commercial datasets. Similarly, Paradinas et al. (2023) applied a Bayesian hierarchical

modelling approach integrating fishery-independent trawl survey data with fishery-dependent trammel net observations in their model accounting for different gear efficiencies. They found that the integrated model clearly improved the niche characterization and spatial prediction of their model species.

Fishery-independent data remains the primary source of occurrence information for species not typically targeted by fisheries, such as marine mammals (see also Kaschner et al. (2012), who highlight the global coverage and biases in cetacean line-transect surveys). However, the high costs, short temporal coverage and low detectability associated with collecting such data have led studies to explore alternative sources of occurrence data.

Acoustic data

Data from passive acoustic monitoring (PAM) has also been increasingly used as occurrence data in SDMs, particularly for vocalizing cetaceans, although they still represent a minority of studies (Pasanisi et al. 2024). This data is commonly used as a complementary occurrence data source for SDMs, typically collected using towed hydrophones during vessel-based surveys or stationary hydrophones deployed in fixed locations. Acoustic methods are particularly valuable for deep-diving cetaceans, for which visual methods might be less efficient, but for which sounds can easily be detected at larger distances than sightings would be possible (Marques et al. 2013, Boisseau et al. 2024). Some studies have explored the utility of acoustic data derived from towed hydrophones in marine SDMs. For instance, Pace et al. (2018) modelled the distribution of sperm whales in the Mediterranean Sea, complementing visual detections with acoustic data from towed hydrophones. Acoustic data from stationary hydrophones has also been used as occurrence data in SDMs. A major advantage of stationary hydrophones is not only their ability to detect species regardless of surface presence but also to potentially provide continuous temporal coverage. This can be particularly advantageous when combined with visual survey data, which typically offer broad spatial but limited temporal coverage. For instance, Frasier et al. (2021) demonstrated that SDMs incorporating acoustic data from fixed hydrophones outperformed models relying solely on visual data for deep-diving cetaceans. Similarly, El-Gabbas et al. (2023) showed that acoustic data can complement visual sighting data in SDMs of blue whales in the Southern Ocean. Despite challenges regarding species identification issues, and lack of knowledge about temporal and spatial aspects of acoustic behavior, acoustic data has proven to be a valuable complement to visual data in SDMs. While most studies currently rely on methods such as data pooling to combine acoustic and visual data, future approaches could emphasize data integration, particularly within Bayesian frameworks that account for the uncertainties inherent in acoustic data. For instance, a recent study from Schliep et al. (2024) used a novel thinned point pattern data fusion approach to integrate acoustic and aerial visual data in modelling North Atlantic right whales. The study demonstrated that this approach

provided improved inference of abundance and distribution, along with performance gains compared to models based on single data sources.

Satellite tracking data

Satellite tracking data has been extensively used as a source of occurrence data in SDMs for particularly marine megafauna species such as cetaceans, sharks, and sea turtles (Whitlock et al. 2016, Hazen et al. 2017, 2018, Ouled-Cheikh et al. 2024). Technological advances in satellite tracking data have fuelled a substantial rise in tracking studies (Katzner and Arlettaz 2019) and they are the fundamental source of occurrence data for animal movement analyses including resource-selection and (integrated) step-selection functions (Avgar et al. 2016, Fieberg et al. 2021). While animal movement studies typically model the movement of individual tracks, applying it to large-scale SDMs may seem

statistically less straightforward due to the typically low number of animals tagged, spatial biases related to tag-on effects, the presence-only nature of this data, and issues related to spatial autocorrelation (O'Toole et al. 2021). However, as more global and meta-analytical studies incorporate tracking data from numerous individual tracks (Queiroz et al. 2019, Hindell et al. 2020) into their movement analyses – and as datasets become more readily accessible through standardization efforts such as Movebank and OBIS-SEAMAP – the potential of tracking data in informing large-scale applications becomes increasingly evident. Some methods, such as random and iterative subsampling of presence and absence data, have been employed to mitigate autocorrelation issues of tracking data in SDMs (Hazen et al. 2018). Nevertheless, future studies should further explore their methodological and statistical applicability in the classical framework of larger-scale (marine) SDMs. Braun et al. (2023) tested

Table 3. Overview of environmental and occurrence data sources used in marine SDMs.

Type	Data source	Description	Examples/References
Environmental data	in situ	Collected via buoys, ships, Argo floats, and sonar-based surveys; provide key environmental covariates like temperature, pH, nutrient concentrations, and dissolved oxygen. Some in-situ collection methods provide depth-specific environmental data.	Forney 2000, Hooker et al. 2001
	satellite remote sensing	Provide broad-scale covariates like SST, chlorophyll-a, and sea ice extent with synoptic coverage and frequent updates.	see Kavanaugh et al. 2021 for review on satellite remote sensing in marine environments
	ocean circulation models	High-resolution outputs from mechanistic models like ROMS, HYCOM, and NEMO; capture dynamic ocean processes (e.g. eddy kinetic energy, current velocities) with high spatiotemporal resolutions.	Becker et al. 2016, Scales et al. 2017, Smith et al. 2021
Most marine SDM studies today derive their environmental data directly from global datasets through portals such as Bio-Oracle, Copernicus Marine Environment Monitoring Service, MARSPEC, Ocean Color Web, or World Ocean Atlas. These datasets often combine the various environmental data sources mentioned above in their products.			
Occurrence data	fishery-dependent data	Catch, bycatch and observer data from (commercial) fisheries are cost-effective, offer broad temporal coverage, but have spatial biases towards fishing areas.	Rezende et al. 2019, Lezama-Ochoa 2020, Karp et al. 2022
	fishery-independent data	Systematic scientific surveys (e.g. via boat, plane or land-based) collect presence and absence data and provide unbiased spatial coverage but are expensive and have limited temporal range.	Pennino et al. 2016, Kanaji et al. 2017, Becker et al. 2020
	opportunistic observations	Data from citizen science such as whale-watching or ecotourism platforms can offer extensive temporal coverage but have spatial biases towards frequently visited regions.	Fernandez et al. 2018, Canales-Cáceres et al. 2023, González García et al. 2023
	passive acoustic monitoring (PAM)	Use fixed or towed hydrophones to detect species-specific vocalizations, can provide high temporal coverage, but have spatial uncertainties.	Taylor et al. 2016, Frasier et al. 2021, Barlow et al. 2022, El-Gabbas et al. 2023
	satellite tracking data	Track marine species movements; valuable for mobile species and foundational data source for movement analyses. Often limited by sample size for large-scale SDMs and autocorrelation challenges.	Hazen et al. 2017, 2018, Braun et al. 2023, Ouled-Cheikh et al. 2024

Integrating the diverse occurrence data sources mentioned above into a single model that retains the strengths and mitigates the biases inherent in individual datasets is a major focus of ongoing research in terrestrial and marine SDMs. Examples of data integration for marine SDMs can be found in: Alglave et al. 2022, Paradinas et al. 2023, Nisi et al. 2024

the utility of various occurrence datasets, including satellite tracking, in their SDMs of blue sharks. All data types produced robust models, with pooled and ensemble approaches offering higher ecological realism. However, since these approaches cannot fully account for uncertainties across datasets (Fletcher et al. 2019, Isaac et al. 2020), future work should focus on integrating satellite tracking data with other occurrence sources. This is increasingly important as more diverse marine occurrence data becomes available.

Alternative data sources

Another source of occurrence data for marine megafauna comes from commercial whale-watching companies. These companies regularly encounter marine mammals, providing observational data that can be systematically collected and used for ecological models. For instance, Fernandez et al. (2018) utilized occurrence data of multiple cetacean species in their SDMs, derived from the MONICET platform (González García et al. 2023; MONICET compiles and standardizes cetacean sighting data in the Azores from various commercial whale-watching operations collected for over more than a decade). Similarly, Canales-Cáceres et al. (2023) utilized opportunistically collected sightings from an ecotourism company for their SDMs of striped dolphins in the Mediterranean Sea. Data from opportunistic whale-watching platforms can offer extensive temporal coverage and numerous sightings, yet they share a common limitation with fishery-dependent data: spatial (and temporal) bias towards areas where cetaceans are frequently and easily observed, and/or where weather conditions are favorable. This preferential sampling presents similar challenges in SDMs as those encountered with fishery-dependent data. Therefore, integrating opportunistic data, from preferential sampling sources, with systematic survey data could present a promising area for future research on marine mammal SDMs. An overview of environmental and occurrence data sources is shown in Table 3.

Conclusions

With the exponential application of SDMs, fuelled by advanced model fitting algorithms and validation techniques, incorporation of biotic predictors, and novel data integration techniques, all supported by ever increasing computational capabilities, their development is accelerating rapidly. In this review, we take a necessary pause to address the essential adaptations required for applying SDMs in marine environments, acknowledging that while advancements in terrestrial SDMs have progressed rapidly, important considerations in marine contexts have been explored but not yet synthesized. Therefore, this review consolidates the key advancements related to the integration of temporal and vertical dimensions in marine SDMs, along with the use of emerging marine data sources. By explicitly synthesizing these components, we provide researchers with guidance on addressing and tailoring these interconnected components to their specific data, ultimately enhancing the applicability of marine SDMs.

We identified numerous studies that have incorporated the temporal dimension into marine SDMs. Given the inherent dynamics of the ocean, where key covariates fluctuate across different spatiotemporal scales, and the fact that many species are highly mobile and interact with these covariates across multiple spatiotemporal scales, moving beyond a static approach is crucial. This is particularly important as many SDM studies aim to provide distribution predictions for dynamic ocean management. We highlight that many studies employ different temporal resolutions in marine SDMs and argue that the choice of an appropriate temporal resolution is a complex task that should be tailored to factors such as the species ecology, the environmental variability of the study area, and the specific objectives of the research and management applications. Further, careful consideration should be given when using broad or fine spatiotemporal resolutions for SDMs. While fine spatiotemporal resolutions can enhance the accuracy of SDMs by capturing fine-scale animal-environment interactions, particularly for highly mobile species, they may not always be appropriate in the classical framework of SDMs that typically operate on larger scales. Therefore, emerging multiscale modelling approaches offer a promising framework for integrating small-scale movement dynamics with large-scale spatial distribution patterns, linking fine-scale animal behaviors with their broader space use (Michelot et al. 2024).

We also reviewed literature on marine SDMs that incorporate the vertical dimension of the ocean in various ways. While many marine SDMs utilize the vertical dimension through static predictors such as depth, others have advanced by testing and incorporating dynamic subsurface covariates. Notably, although studies are still limited, important progress has been made in 2.5D and 3D SDMs, particularly for species inhabiting deeper ocean layers. These models link the depth at which species are recorded with corresponding environmental covariates at that specific depth. This provides a more accurate representation of their habitat, which is a logical and necessary progression in marine SDMs of those species.

Outstanding questions remain regarding if and how the vertical dimension should be incorporated when modelling species that exhibit strong vertical movements, such as deep-diving cetaceans. These species dive to great depths in pursuit of prey, yet traditional SDMs treat their spatial presence in only two dimensions. This raises the question of whether a 3D SDM could provide more accurate representations. However, considering that their occurrence at depth is time-dependent and can only be obtained through bio-logging devices, combined with the limited availability of environmental data at depth, a 3D approach presents challenges and may not be practical for large-scale SDM applications. When employing 2D approaches – coupling surface covariates with species that inhabit deeper water layers or exhibit significant vertical movements – careful consideration is needed. This is particularly important during model extrapolation, as the model may function well in the geographic space but may be compromised within the environmental space (Elith and Leathwick 2009). If a 3D approach is not feasible due to unavailability

of data or uncertainty of vertical space use of model species, incorporating subsurface covariates in a 2D framework might be a good alternative (Abrahms et al. 2019b). Since dynamic subsurface covariates are often sourced from ocean circulation models that are not always available, it is important to note that surface features can sometimes indicate subsurface processes (Wilson and Coles 2005). For example, surface oceanic fronts are linked to subsurface nutrient upwelling and prey aggregation (Scales et al. 2014). Incorporating surface-derived covariates that reflect these subsurface dynamics could then be a practical alternative, especially when direct subsurface data are unavailable. Nevertheless, to explicitly account for vertical movements for SDMs, future multiscale approaches that integrate fine-scale animal movement with larger-scale distribution (Michelot et al. 2019) could present a promising direction for research. While we have highlighted the potential of marine data sources, such as environmental data from numerical ocean circulation models and occurrence data obtained through collaborative efforts, challenges persist. Firstly, the availability and quality of environmental data are inconsistent across regions, particularly in deeper and more remote areas. Although many environmental data sources provide fine spatiotemporal resolutions, particularly those derived from ocean circulation models, such data are not always available for certain regions. Further, in regions and depths where in situ data is sparse, marine SDMs often depend on interpolated data, which can introduce uncertainties into the model. Also, the quality and quantity of occurrence data in marine environments is often a limiting factor for SDMs. Unlike terrestrial environments, where species occurrence data is relatively common, marine occurrence data is often sparse, particularly for species not targeted by fisheries. Moreover, data from fishery-dependent sources are often biased towards commercially valuable areas, which can lead to a biased representation of marine species distributions. Therefore, addressing these challenges through data integration techniques, that account for preferential sampling, is particularly promising for marine environments and could be a key focus for future advancements in marine SDMs.

While we have discussed three areas of progress in this review, there are other challenges that persist, particularly regarding more technical modelling aspects. For instance, inter- and extrapolation are current challenges that are shared with terrestrial systems but may be amplified in dynamic marine environments especially in remote areas and when data is sparse (Yates et al. 2018, Bouchet et al. 2020). These can introduce substantial risks when predictions extend beyond sampled conditions, where assumptions like niche conservatism (Strubbe et al. 2013) might break down. For instance, Guillaumot et al. (2020) found that for their models for benthic species in the Southern Ocean up to 78% of the projection area was extrapolation. For this, analytical tools to evaluate the reliability of predictions, especially under extrapolation scenarios, have been developed and tested such as multivariate environmental similarity surface (MESS) (Elith et al. 2010, Guillaumot et al. 2019, 2020), mobility-oriented parity (MOP) (Owens et al. 2013, Cobos et al.

2024) and extrapolation detection (ExDet) (Mesgaran et al. 2014, Bouchet et al. 2019, 2020). These tools can assist to assess uncertainty in SDMs when models are applied beyond their training data. Assessing and reporting uncertainty from inter- and extrapolation, as well as from data deficiencies and model parameters (Miller et al. 2022), is crucial for marine SDMs particularly when they are used in management and conservation contexts (Redfern et al. 2017). Therefore, we refer to the marine SDM framework that Robinson et al. (2017) developed, where uncertainty reporting is an integral part of the model-building process. Looking ahead, advances in data integration procedures represent promising directions to address uncertainty and extrapolation challenges in marine SDMs. In particular, approaches that jointly model multiple observation processes – such as structured surveys and opportunistic data – can expand the coverage of geographic and environmental space, thereby improving inference in under-sampled marine regions.

Outlook

We are encouraged by the increasing recognition of the unique challenges that the marine environment presents for SDMs, particularly regarding incorporating the temporal and vertical dimensions. While temporal and vertical dynamics pose similar challenges for marine SDMs – resolution mismatches, data gaps, and extrapolation risks – they also offer shared opportunities. As data and methods improve, future marine models should aim to take into account both dimensions, capturing the full spatiotemporal complexity of species habitat use in marine environments. Lastly, we observe a strong methodological overlap between marine SDMs and terrestrial SDMs. For instance, both fields address high environmental heterogeneity, such as microclimates in terrestrial systems (Lembrechts et al. 2019) and dynamic oceanographic features in marine environments. Further, dynamic SDMs, as discussed here, also play a major role in terrestrial avian SDMs, since many bird species, similar to marine species, exhibit high mobility. This overlap and common challenges highlight a valuable opportunity for greater dialogue and collaboration between researchers in both fields.

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Author contributions

Moritz Klaassen: Conceptualization (lead); Funding acquisition (lead); Investigation (lead); Visualization (lead); Writing – original draft (lead); Writing – review and editing (lead). **Tiago A. Marques:** Conceptualization (equal); Supervision (equal); Writing – review and editing (equal). **Filipe Alves:** Conceptualization (equal); Funding acquisition (equal); Supervision (equal); Writing – review and editing (equal). **Marc Fernandez:** Conceptualization (lead); Funding acquisition (equal); Supervision (lead); Writing – review and editing (equal);

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Data availability statement

Data sharing is not applicable to this article as no new data were created or analyzed in this study.

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